SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security COURSE CODE: CSA5117

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Akshitha S/192521396 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Akshitha.S

Signature of the student:

i) Effect on Data structure and security level

Substitution:

* changes the actual symbols in the message.
* The length of the message generally remains the same.
* Letter frequency patterns may remain, especially in simple substitution.
* Therefore, simple substitution ciphers can be vulnerable to frequency analysis.

Transposition:

* Does not change the actual characters.
* Only changes their positions.
* The frequency of characters remains exactly the same.
* If the transposition pattern is discovered, the plaintext can be recovered.

ii) Why is More Vulnerable and Why?

Simple substitution techniques are generally more vulnerable because frequency analysis can reveal repeated letters and common patterns.

For example, in English, E is one of the most

frequently occurring letters. In a simple substitution cipher, its frequency pattern is still preserved, although the symbol representing E is changed.

Transposition also has weaknesses because character frequencies remain unchanged, but combining substitution + transposition provides stronger protection than using either alone.

(4) Concept Map - steganography

a) Relationships Between the Concepts

1. Data Hiding

Data Hiding means Concealing secret information inside another cover medium, such as image, audio file, video or text.

Example:

A secret message is hidden inside the pixels of an image without making the image look significantly different.

2. Watermarking

Watermarking embeds information into digital media to identify ownership,

Copyright or authenticity.

Example:

A photographer embeds a copyright watermark into a digital photograph.

3. Survivability

Survivability means the hidden or embedded information should remain detectable even after operations such as:

* Compression
* Resizing
* Editing
* Transmission
* Format Conversion

Relationship

Data Hiding

↓

Conceals information

↓

watermarking

↓

protects ownership/Integrity

↓

Survivability

↓

Ensures embedded information survives attacks

b) Contribution to secure communication.

Data Hiding:

Keeps the existence of secret communication hidden.

Watermarking:

provides copyright protection, authentication and integrity.

Survivability:

Helps hidden information remain available after modification or attack.

How they work together

For secure communication, a sender can hide information inside an innocent-looking image. A watermark can identify the legitimate sender or owner, while survivabil ensures the embedded information remains usable even if the image undergoes compression or modification.

c. Real-World Application for each

1. Data Hiding - Secret Communication

A confidential message can be hidden inside an image and sent through a normal communication channel. the

existence of the message is difficult to notice.

Application : Secure / private communication

2. Watermarking - Copyright protection
Digital photographs, movies and documents can contain an invisible watermark identifying their owner.

Application: Copyright and ownership protection.

3. Survivability - Digital media protection
A watermark embedded in an image can be designed to survive JPEG compression, resizing or minor editing.

Application:
protecting digital media from unauthorized use.